Doddi Sarvan Naidu

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	9.00
Sri chaitanya	Class XII (Board of Intermediate Education, Andhra Pradesh)	2024	97.80%
Sri chaitanya	Class X (Board of Secondary Education, Andhra Pradesh)	2022	96.17%

SCHOLASTIC ACHIEVEMENTS

- **IMC Prosperity 4 (Global Quant Trading Competition):** Achieved **AIR 3 (Worldwide Rank 36)** in the Penultimate Round and **AIR 47** in the Ultimate Round.
- **ICPC Preliminary Round:** Achieved **Rank 173**.
- 800 rank in iicpc quantfest out of 10000+ participants

SKILLS

- **Languages**: C, C++, Java, JavaScript, Python, SQL (MySQL, PostgreSQL), OCaml, RISC-V Assembly
- **Technologies & Tools:** Node.js, Git, Linux, Docker, Solid Edge
- **Frameworks**: Express.js

RELEVANT COURSES

- **Computer Science**: Introduction to Programming, Software Development Fundamentals, Data Structures, Algorithms, Computer Architecture, Operating Systems, Database Management Systems, Compilers I, Theory of Computation
- **Mathematics**: Discrete Mathematics, Calculus, Linear Algebra, Probability & Random Variables, Transform Techniques
- **Engineering Fundamentals**: Digital Circuits ,Basic Electrical Engineering

PROJECTS

Multi staged pipelined RISC-V CPU Simulator (C++) GitHub

Computer architecture course

- Extended an existing single-cycle RISC-V simulator into a configurable architecture simulator with six execution modes, ranging from single-cycle to pipelined execution with hazard handling, forwarding and branch prediction.
- Implemented complete 5-stage pipeline functionality (IF, ID, EX, MEM, WB), including pipeline registers, control signal propagation, hazard detection, operand forwarding, and pipeline flushing for data and control hazards.
- Developed both static (BTFNT) and dynamic n-bit branch predictors with prediction accuracy tracking, along with configurable pipeline behavior for comparative performance analysis.
- Built a configurable set-associative cache simulator supporting write-back, write-allocate, LRU replacement, variable associativity, block size, and detailed cache performance statistics.
- Added performance instrumentation including cycle counts, pipeline bubble analysis, cache hit/miss statistics, branch prediction accuracy, per-stage latency profiling, register dumps, and interactive undo/redo execution.

xv6 Kernel Extensions (C, xv6) GitHub

Operating systems course

- Extended the xv6 kernel with custom system calls and a Multi-Level Feedback Queue (MLFQ) scheduler supporting

priority-based preemption, time-slice management, and starvation prevention.
- Designed and implemented a disk-backed virtual memory subsystem with page eviction, swap-space allocation, swap-in/swap-out handling, and page-fault driven memory recovery to support execution beyond available physical memory.
- Developed configurable disk scheduling algorithms for swap I/O management and integrated RAID 0, RAID 1, and RAID 5 based multi-disk storage policies for swap block allocation and recovery.
- Implemented kernel-level statistics collection for memory, swap, and disk operations, enabling detailed analysis of I/O behavior, paging activity, and system performance under different workloads.
- Evaluated the impact of scheduling, swapping, and storage configurations through workload-based performance measurements and system instrumentation.

Infinite Precision Arithmetic calculator (Java) GitHub

software fundamentals course

- Developed a custom arbitrary-precision arithmetic library supporting integer and floating-point computations beyond native data type limits without using built-in big number libraries.
- Implemented AInteger and AFloat classes using string-based representations with support for addition, subtraction, multiplication, division, comparison, parsing, and signed arithmetic.
- Designed algorithms for decimal alignment, normalization, quotient/remainder computation, and precision management to ensure accurate operations on extremely large and high-precision numbers.

Core-ML Exception Handling Extension (Ocaml) GitHub

- Added raise and try-with exception handling to MiniML by extending the lexer, parser, AST, type checker, compiler, and abstract machine.
- Implemented DivisionByZero and GenericException with support for raising, catching, and handling exceptions during program execution.
- Updated the compiler and runtime to generate and execute exception-handling instructions while preserving the language's existing functionality.

EXTRACURRICULAR

- **Competitive Programming:** Achieved a peak **Codeforces** rating of **1535**.
- **Problem Solving:** Solved **600+** algorithmic problems across multiple coding platforms.
- Member of the winning team in the **Nexus Competition** conducted during **ELAN & Vision, IIT Hyderabad**.
- **Karate:** Awarded the **Brown Belt** in Karate